Modeling Flexibility Services in Energy Communities

Bibliographic project – Master E2SD

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1 Introduction

1.1 Context

We are now well aware that we need to significantly reduce our greenhouse gas emissions to limit global warming. Electrifying as many sectors as possible and decarbonizing electricity production are two major ways to achieve this goal. However, the renewable energy sources used to produce clean electricity, such as solar and wind power, are intermittent and require much more space than traditional power plants. This has led to growing interest in decentralized electricity production, as it can spread generation over a larger area to reduce intermittency issues and take advantage of the many available surfaces. For example, urban areas have numerous unused surfaces that could be used to install solar panels, such as rooftops and building facades.

By installing renewable energy sources at home or in local areas—particularly solar panels—a former consumer can become a "prosumer," meaning they both consume and produce electricity within the grid. We can then distinguish two cases of production: either the prosumer injects power into the grid, or the prosumer self-consumes the produced power. Depending on policies and grid fees, self-consumption is often much more attractive for prosumers than injecting power into the grid.

To maximize self-consumption, we can generally consider two main approaches. The first is to install energy storage systems to store excess production for later use. However, storage systems are still quite expensive. The second approach is to adapt consumption to production, which can be difficult to achieve at the individual level. This is where local energy communities (LECs) become useful. By grouping prosumers together, they can share production and consumption, thereby maximizing self-consumption at the community level. Of course, LECs can also benefit from storage systems. When a community is built using only renewable energy sources and storage systems, we refer to it as a "renewable energy community" (REC).

1.2 Local energy communities

This study will focus on LECs and, more specifically, on RECs in Europe, as this concept depends heavily on local regulations. Of course, LECs also exist in many other regions of the world, and the challenges they face are likely quite similar.

Energy communities were introduced in the EU in 2016 under the Clean Energy Package, particularly through the Renewable Energy Directive (REDII), and have since grown significantly. The adoption of legislative measures such as REDII and the Directive on Common Rules for the Internal Electricity Market (EMDII) provided the first formal definitions of energy communities in European law. RECs typically connect to public energy networks operated by local DSOs, but the 2019/944/CE Directive promotes closed distribution systems (CDSs). These are intended for the exclusive use of REC members and are expected to organize exchanges within the REC. The European directive does not impose restrictions on how RECs may distribute profits among participants, so this responsibility would also fall to the CDSs.

In most European countries, energy communities are limited in geographic scope and number of participants, based on the principle that they should be connected to the same feeder. Nevertheless, there is considerable diversity in the size of energy communities, ranging from a few households at low voltage to industrial parks at medium voltage.

The main goal of RECs is to involve communities in the energy transition by supporting investment in renewable energy infrastructure and by encouraging the integration and use of flexibility services. Therefore, social acceptance and community engagement are crucial aspects to ensure their success. Typically, the first objective of RECs is to maximize the profit of their members, which can be achieved through optimization processes. This optimization generates financial gains for the community, which should then be distributed among its members. Several

methods can be used for this purpose, but they may not be equally fair to all participants. Another important aspect to consider is the impact of RECs on the electrical grid. By adapting their consumption to their production, RECs can provide flexibility to the grid and help manage demand. However, if not properly managed, they could also create stability issues.

1.3 Organisation of the bibliographic study

The goal of this document is to provide an overview of the current state of the art regarding renewable energy communities using flexibility services. It is organized into three main sections. The first section focuses on research related to financial gains and equitable distribution among community members. The second section examines the impact of RECs on the electrical grid, including flexibility and stability issues. Finally, the third section addresses the social dimensions of RECs, such as community engagement and social acceptance.

2 Optimization and equitable distribution

In this section, the methods to maximize the financial gains and ecological benefits of RECs will be discussed. Then, the different ways to distribute the gains among the members of the community will be presented.

2.1 Presentation of the optimization problems

In a REC, there are some usual common parameters. Each member of the community has its own consumption and production profile. Furthermore, some members can have a battery storage system. And as we often want to optimize the self-consumption, the members have a certain number of flexible loads with different kind of flexibility.

For members that are small consumers at the low voltage level, we can identify different types of standard flexible loads.In [1], the considered flexibilities are of three types :

- White goods (washing machine, dryer, dishwasher...) with a fixed time of use but flexible starting time
- Electric vehicle with a charging power that can be controlled but with constraint on the state of charge at departure time
- Thermal loads that are fully flexible loads

In [1], the way the different flexibilities are managed is associated with a discomfort price in the objective function.

[2] consider two main categories of flexible loads : shiftable loads and sheddable loads. Shiftable load are the same as white goods while sheddable loads can be reduced in power for a certain amount of time. So in the previous example, thermal loads and electric vehicles can be considered as sheddable loads.

Finally, in [3], they modelize loads in a house for four categories to construct optimization models :

- Non-flexible loads, they profile is fixed and cannot be changed
- Shiftable loads, as mentioned before
- All or nothing loads, that can be turned on or off at any time
- Electric vehicle loads, here they are considered "All or nothing" when they are at home

By considering all of these flexible loads model it is possible to build optimization models that take into account all the different flexibilities of the members of the REC.

From here on, several parameters can be optimized depending on the goals of the REC. Usually, the cost is minimized as in [4–6], a discomfort cost as in [1] can also be added to the objective function. In [7], the goal is to maximize the robustness of a community that would be disconnected from the main grid. Finally, the ecological impact can also be minimized as in [3] by minimizing the CO2 emissions associated with the electricity consumption of the community. In [5], they propose a model that take into account demand response requests from the distribution system operator (DSO) in order to provide services to the grid.

In the case where the REC is disconnected from the main grid, what is studied is the operability of the community regarding the uncertainties of production and consumption. [7, 8] both uses robust optimization with a min-max-min approach to ensure the operability of the community in the worst-case scenario.

Several approaches can be used in order to optimize the parameters of the community. In [1,4] a centralized approach is considered. This means that a central entity has access to all the data of the community and can optimize the parameters for the whole community. Of course, this will reach the best case scenario as all the information is available. However, this can lead to privacy issues and higher computation time.

To avoid this, [6] proposes different approaches with different levels of data communication. The first approach is the fully centralized one. Then they propose a method where the aggregator communicates the prices to the agents, then the agents propose a first optimal strategy to communicate their surplus and deficits to the aggregator. And then a loop is initiated between the agents and the aggregator until convergence in order to optimize the battery usage. Finally, a fully decentralized, named "selfish", approach is proposed.

2.2 Allocation methods within the community

With the optimization process, the REC can achieve an optimal planning strategy for exchanging energy within the community and with the main grid. This strategy will lead to overall financial gains for the community.

In order to distribute these gains, we can use several methods : pricing mechanisms in the community, allocation functions that directly distribute the gains and game-theoretic approaches. Before exploring these methods, the tariff for a single prosumer will be presented.

In this section, the community is made up of a set of agent that represents the community members. Here, each agent is a prosumer, it is associated with a production and a consumption profile that are optimized in the operation phase. A member that does have energy sources will have a null production profile.

2.2.1 Single prosumer

Depending on the contract of the prosumer, they will pay a price $p_{buy}(t)$ for each kilowatt-hour consumed. This price might depend on the time of the day. The same way a price for selling is defined $p_{sell}(t)$. Usually, there will also be a subscription fee that depends on the maximum power the prosumer is allowed to use. The subscription fee will not be discussed here as the community will mainly act on the energy prices.

A total cost can then be defined considering the power consumed P_{cons} and produced P_{prod}.

$$Cost = \sum_{t=T_0}^{T_{end}} p_{buy}(t)P_{cons}(t) - p_{sell}(t)P_{prod}(t) \tag{1}$$

2.2.2 Pricing mechanisms

A common way for allocating the gains is to use pricing mechanisms that will determine the price of energy exchanged within the community. In this case, a local market is created, the prosumers will then buy/sell energy from/to the local market.

Mid-market rate (MMR)

This method uses the feed-in tariff, which guarantees energy producers a fixed price for renewable electricity fed into the grid under long-term contracts. An important property is to have a fixed reference for the selling price in this method.

The MMR pricing strategy [4] is an average price between the time-of-use and the feed-in tariff. So at each instant we distinguish 3 cases.

Production of agent i is greater than its consumption, and the total community balance is positive, then :

$$price_i = p_{mid} = \frac{p_{buy} + p_{sell}}{2}$$

Production of agent i is less than its consumption, and the total community balance is positive, then :

$$price_i = \frac{p_{mid} \cdot \sum_i P_{prod_i} + p_{buy} \cdot |\sum_i P_{cons_i} - P_{prod_i}|}{\sum_i P_{cons_i}}$$

Where P_{prod_i} and P_{cons_i} are respectively the production and consumption of agent i.

Production of agent i is less than its consumption, and the total community balance is negative, then :

$$price_i = \frac{p_{mid} \cdot \sum_i P_{prod_i} + p_{sell} \cdot |\sum_i P_{cons_i} - P_{prod_i}|}{\sum_i P_{cons_i}}$$

Supply-demand rate (SDR)

The SDR pricing strategy [4] sets the internal community price according to a ratio between the community net generation and net demand. At each instant, we have :

$$rate = \frac{\sum_i P_{prod_i}}{\sum_i P_{cons_i}}$$

$$price = \begin{cases} p_{sell} & \text{if } rate > 1 \\ p_{buy} & \text{if } rate = 0 \\ p_{sell} rate + (1 - rate) & \text{if } 0 < rate \leq 1 \end{cases}$$

The price here, is defined as being the same for everyone in the community.

Peer-to-peer transactions

In [9], they propose a peer-to-peer pricing mechanism where agents can build bilateral contracts with each other to trade energy. The price of energy is determined by a market mechanism where agents can propose bids and offers. In this study, the agents are paired randomly, but other pairing methods can be used.

2.2.3 Total profit allocation functions

Another way of allocating the gains within the community is to only distribute the total profit of the community without using pricing mechanisms. Different methods can be used to do so [4,9].

Equal Allocation of Non-Separable Values (EANSV)

This method [9] is based on comparing the individual initial cost without being part of the community and the collective cost of the community. For this it determines the non-separable values that are the values could not be attributed to a single agent. So this method attribute as a cost to each agent its separable cost plus an equal share of the non-separable cost. It can be expressed as follows for an agent i :

$$C_i^{EANSV} = C_i^{ind} + \frac{C_{community} - \sum_i C_i^{ind}}{N}$$

Where C_i^{ind} is the individual cost of agent i without being part of the community, $C_{community}$ is the total cost of the community and N is the number of agents in the community.

Proportional Allocation (PA)

This method consist of distributing an equal proportional saving to each agent. It can be expressed as follows for an agent i :

$$C_i^{PA} = C_i^{ind} + \frac{|C_i^{ind}|}{\sum_i |C_i^{ind}|}(C_{community} - \sum_i C_i^{ind})$$

2.2.4 Game-theoretic approaches

This allocation problem can also be studied as a cooperative game. We can then use game theory for computing the distribution. In [3,4], they propose 2 other methods based on game-theoretic approaches : the Shapley value and the Nucleolus.

Those two methods uses the principle of coalitions between agents. A coalition S is a subset of N, the set of all agents. The value of a coalition S is given by a characteristic function $v(S)$ that gives the total profit of the coalition S.

Shapley value

The Shapley value [3,4] is defined as follows for an agent i :

$$\phi_i(v) = \frac{1}{n} \sum_{S \subseteq N \setminus \{i\}} \binom{n-1}{|S|}^{-1} [v(S \cup \{i\}) - v(S)]$$

with n being the number of element in N, $|S|$ is the number of element in set S, and $\binom{n-1}{|S|}$ is a binomial coefficient.

This formula can be interpreted as the mean values of the contribution of agent i in each possible coalition without agent i.

This method is considered to be fair as it takes into account the marginal contribution of each agent to all possible coalitions. However, it can be quite complex to compute as the computation time grows exponentially with the number of agents.

Nucleolus

The nucleolus [3,4] of a cooperative game is the solution that maximizes the smallest excess of a coalition. The excess of a coalition S for a given allocation x is defined as :

$$e(S, x) = v(S) - \sum_{i \in S} x_i$$

So the nucleolus can be obtained by solving a serie of linear programming problems that minimize the maximum excess of all coalitions. It is also often simplified using only the prenucleolus, as the nucleolus is quite heavy to compute.

2.2.5 Other methods

Other methods have been proposed such as optimization based allocation. Those can be used to propose strategies that answer specific needs of the community. For example, in [9], they propose an optimization framework that aims to take into account the individual preferences of the agents regarding their engagement in the community for ecological purposes.

Dynamic pricing mechanisms using bids and offers can also be used to determine the price of energy exchanged within the community. In this case, several bids strategies can be used.

2.3 Fairness indexes and equitable distribution

The shapley value and the nucleolus are two methods that aim to provide a fair distribution of the gains among the members of the community. However, there are several ways to define fairness. In [4,9–11], they identify different fairness index to evaluate the fairness of a distribution method.

2.3.1 Jain's index

Jain's index [4, 9–11] is one of the most common fairness index. It measures the equality of the distribution and is defined as follows :

$$J(x) = \frac{(\sum_{i=1}^{N} x_i)^2}{N \cdot \sum_{i=1}^{N} x_i^2}$$

Where x is the vector of gain allocations to each agent. The index ranges from 1/N (worst case) to 1 (best case).

2.3.2 Participation index

The participation index [10] measures the proportion of agents that benefit from the community. It is defined as follows :

$$PI(x) = \frac{N_{CostReduced}}{N}$$

Where $N_{CostReduced}$ is the number of agents that have a lower cost being part of the community than being alone.

This index can be used to make sure that the distribution method does not disadvantage too many participants.

2.3.3 MinMax index

The MinMax index [9] measures the ratio between the maximum and minimum allocation among the agents. It aims to evaluate the equality of the distribution in the same way as Jain's index. It is defined as follows :

$$MM(x) = \frac{min_i x_i}{max_i x_i}$$

2.3.4 Quality of Experience (QoE) index

This index compare the standard deviation to its maximum possible value [9]. So the aim is still the same as for the Jain's and MinMax index. It is defined as follows :

$$QoE(x) = 1 - \frac{\sigma(x)}{max(x_i) - min(x_i)}$$

2.3.5 Comparison to Shapley value

The last index measure the difference between the obtained allocation and the Shapley value allocation [4, 9–11]. It is defined as follows :

$$\Delta = 1 - \sum_{i=1}^{N} \frac{x_i}{\sum_{k=1}^{N} x_k} - \frac{x_i^{SV}}{\sum_{k=1}^{N} x_k^{SV}}$$

Where x^{SV} is the allocation given by the Shapley value method. The Shapley value is deemed as fair as it calculates the marginal contribution of each agent to all possible coalitions.

Using those different fairness indexes, it is possible to evaluate the fairness of a distribution method within a community. This can help choose the most suitable method for a given community depending on its priorities.

3 Impact on the electrical grid

3.1 Flexibility and demand-side management

Flexibilities on the consumer side can help the grid for its stability. Thus, there is now more and more incentives using tariffs for consumers to provide flexibility to the grid. In RECs, those flexibilities are even more important. As the goal of RECs is to maximize self-consumption, and the renewable energy sources, mainly solar panels, have a high variability in their production, it is important to be able to adapt the consumption to the production. This is even more true when the REC is supposed to work in islanded mode.

One could think that this flexibility of the RECs could help the grid by providing demand-side management services. However, in [12], they show that the impact of RECs on the grid is not always positive. A bad management of the REC could lead to production peaks and to congestions in the grid. Therefore, it is essential to manage the flexibility services within the REC effectively to create as few instabilities in the grid as possible or even to provide services to it.

The type of flexibility has been listed in section 2.1.

3.2 Impact on internal stability of the community

In [7], they study the operability of a REC that would be disconnected from the main grid in order to study stability in the worst case scenario. Using a robust optimization approach, the amount of flexibility of each user is specified by each agent and taken into account as a constraint in the optimization problem. With this method, they show that it is possible to ensure the operability of the REC in the worst case scenario, even if this leads to a higher cost for the community.

They also studied their optimization method without being in island mode. In this case, they show that it reduces the costs while optimizing the use of local renewable energy sources. So there is a trade-off to find between ensuring the stability of the REC and minimizing the costs.

3.3 Impact on overall grid stability

In [12], they study the impact of RECs on the overall grid stability. They compare two scenarios. In the first one the prosumers are not organized within a REC, while in the second one they are, and their exchanges are optimized to maximize self-consumption. In each case, the number of flexibilities remain the same.

They show that for some days, the REC can help to reduce the voltage deviations. But for other days, it is the opposite. The REC can create voltage deviations that were not present in the first scenario and at times when it is not classically expected. The authors specify that this is mostly true when the RECs are not served by the same feeders or if they are located at a low or medium voltage node in the grid. So the way RECs are built and operated can have a significant impact on the overall grid stability.

In [5], they model a REC where the DSO can send demand response requests (DR) to the REC in order to provide services to the grid. The DR requests are associated with a time horizon and a reward. With the rewards fixed as in their study, they show that the REC can provide services to the grid in order to maximize its profit. Hence, the DSO could use DR requests to ensure the stability of the grid.

4 Social Dimensions

4.1 Community engagement

One of the objective of RECs as defined by the European Union is to involve the community in their energy management and in the energy transition. In a more concrete manner, this engagement is meant to provide the needed flexibilities for the community as well as the investment such as PV panels and battery storage. Thus, community engagement is a very important aspect to consider in order for RECs to be successful. [3] describes 3 main sociological profiles regarding the engagement in a REC :

- The "environmentalist" profile, they are motivated by ecological concerns
- The "economic" profile, they are motivated by financial gains
- The "independent" profile, they are motivated by the idea of being autonomous regarding energy

Each member of the community can share more or less one of these profiles. In order to engage the community, it is important to take into account those different profiles while operating the REC. For example, some members might be more willing to accept a higher cost if it leads to a lower ecological impact.

Taking this into account, the optimization process can be adapted using different metrics as objectives. In [3], they consider for example self-consumption rate, CO2 emissions and cost as metrics to optimize. The fairness index presented previously are also important to consider depending on the profiles of the members of the community.

[2] developed a serious game in order to study the flexibility of individuals in a REC. The participants are separated into 4 groups, one with community, another without and with or without peer-to-peer (P2P) pressure. Throughout the game, each participant will have the choice to accept or not load shifting and load shedding. The study lasted for 2 weeks, in the first week, there is not too much incentive from the game to accept flexibility. In the second week, a score is now given to each participant that depends on their actions.

Between the participant in a community with P2P pressure and the participant alone without P2P pressure, there is a 58% increase in acceptance of the suggested flexibility. But this study concludes that being part in a community and having P2P pressure is the only way to significantly increase the acceptance of flexibilities.

4.2 Social acceptance

[10] presents a review of the literature on the fairness in local energy systems. They show that fairness withing a community is a key factor in order to put in place systems such as RECs.

Furthermore, the concept of fairness is quite subjective and vary depending on the studies that want to take it into account.

They identify 3 main dimensions of fairness :

- Equality, each participant should receive the same amount of benefits
- Min-max fairness, the allocation should maximize the minimum benefit received by a participant
- Meritocracy, the allocation should be proportional to the contribution of each participant

Looking back at the different distribution methods presented previously, the rule based methods such as pricing mechanisms focus on meritocracy. While the redistribution methods can choose more to focus on the aspect they want to prioritize. Then the different indexes seen in 2.3 can be used in order to check the fairness of the final distribution.

As mentioned in [9], some of the distribution method presented can be quite hard to understand for an average member of a community. This problem of explainability can be a challenge that needs to be addressed for social acceptance.

We could also think that this is also true for the optimization processes. Indeed, in order to optimize the operation of a REC, several algorithms can be used. Those will result in decisions that will impact the members of the community and maybe the DSO. This can be particularly important when metaheuristic algorithms are used as they are not proven to converge toward the optimal solution. In [13], they propose a review on the different ways to make those algorithms more explainable and to justify that they will converge toward a good solution.

5 Conclusion

Key takeaways

This article presented an overview of the state of the art regarding local energy communities. In Europe, a legal framework around energy communities has been developed to encourage the installation of renewable energy sources. These sources are often PV panels, which have very high variability. RECs can therefore be used first to engage a community in installing the infrastructure, and then to optimize the usage of the energy produced.

Different optimization frameworks have been developed to optimize various objectives. Economic gain and self-consumption are often the two main values optimized, but some articles also study CO2 emissions or the stability of the local grid under worst-case scenarios.

This can be done through centralized methodologies or more decentralized ones. This choice is important regarding the complexity of the algorithms used, as the more centralized the method, the more computation it may require. It can also impact community members, as a centralized method will need precise data on their everyday life, which may raise privacy concerns.

Another key aspect of this optimization process is how the benefits from the community are distributed among its members. Different methods exist for this as well. In the literature, we find dynamic pricing mechanisms, distribution functions, and game-theoretic approaches using Shapley value and Nucleolus. We also find optimization approaches that are more specific to the studied community. The choice of one method over another will lead to a more or less fair distribution.

To assess the fairness of each method, different indices have been presented, which can be used depending on the chosen definition of fairness. Indeed, three main dimensions of fairness coexist: equality, min-max fairness, and meritocracy.

The choice of the optimization framework will greatly influence the operation of the community and might affect community engagement. That is why, in this field, it is necessary to

take into account a sociological aspect. Three main sociological profiles have been studied: the environmentalist, the economic, and the independent. Community members can be modeled using these profiles. By studying the profiles within a community, we can attempt to find an appropriate optimization framework.

Furthermore, the way the community is optimized could influence the stability of the grid. Studies show that a REC optimized for self-consumption can create line congestion depending on how the community is structured. Finally, the social engagement of community members has been studied through a serious game. The results show that peer pressure can greatly help members accept the use of more flexibility.

Future Work

Looking at the literature, it seems there is still a lack of studies exploring the impact of developing local communities on the electrical grid. We can assume that RECs could be a way to install local renewable power sources without overly affecting the grid, if operated properly, but this needs further study. To study larger communities or systems of several communities, the optimization framework needs to be scalable. There are already studies exploring this aspect, but further work is needed to cover every aspect of optimization, from operation to fair distribution. In particular, game-theoretic approaches are difficult to scale.

Finally, all of these optimization processes are still very theoretical and lack real-world applications. Case studies would be essential to prove the effectiveness of the presented methodologies and to support the installation of more renewable energy sources.

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